



Question(s): 1/2

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CONTRIBUTION**Source:** India**Title:** Addressing, naming & identification mechanism for fixed and wireless users served by converged core IMS network**Contact:** Smita Pawar
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Tel: +912279659281
E-mail: yogesh.naik@ril.com**Abstract:** This contribution provides the baseline text for using domain name, private identities and public identities for IMS based services in different geographic areas.**1 Scope**

The method for deriving domain name, private identities, and public identities for offering multimedia services using IP Multimedia Subsystem (IMS) over different geographical regions is described in this draft guideline.

2 References

- [IETF RFC 3261]: SIP: Session Initiation Protocol.
[3GPP TS 23.003]: Numbering, addressing and identification.
[3GPP TS 23.228]: IP Multimedia Subsystem (IMS); Stage 2.
[3GPP TS 24.228]: IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3
[ITU-T E.164]: Recommendation ITU-T E.164 (2010), The international public telecommunication numbering plan.

3 Definitions**3.1 Terms defined elsewhere**

[TBD]

3.2 Terms defined in this Recommendation

[TBD]

4 Abbreviations and acronyms

3GPP	Third Generation Partnership Project
CLI	Calling Line Identity
ENUM	Electronic Number Mapping
HSS	Home Subscriber Server
IMPI	IP Multimedia Private Identity
IMPU	IP Multimedia Public Identity
IMS	IP Multimedia Subsystem
IMSI	International Mobile Subscriber Identity
IP	Internet Protocol
IP-PBX	Internet Protocol Private Branch Exchange
MCC	Mobile Country Code
MSISDN	Mobile Station International Subscriber Directory Number
MSO	Multiple System Operator
NAI	Network Access Identifier
NMNC	Mobile Network Code
PLMN	Public Land Mobile Network
PSTN	Public Switched Telephone Network
RURI	Request-URI
SIP	Session Initiation Protocol
TCO	Total Cost of Ownership
URI	Uniform Resource Identifier

5 Conventions

[TBD]

6 Overview

The Third Generation Partnership Project (3GPP) developed the IP Multimedia Subsystem (IMS) to define an infrastructure for all-IP communication services. Initially, IMS was designed to meet the requirements of mobile operators by combining the IP technology of the internet with the wireless and mobility services of mobile telephony networks. However, it was later extended to include fixed networks as well. The 3GPP chose to adopt Session Initiation Protocol (SIP) as the signalling protocol for session establishment and control in IMS, which has paved the way for a close integration of the fixed, mobile, and Internet worlds.

The Session Initiation Protocol (SIP) signaling protocol is widely used for initiating, modifying, and terminating multimedia communication sessions, such as voice and video calls over IP networks. With SIP, multimedia communications, whether between IMS users or between Internet users, can apply the same control protocol, thus truly realizing convergence of communications network and Internet. In SIP, numbering refers to the addressing and identification of entities involved in the communication.

7 Identities in SIP

Two types of identities are used in SIP to address participants and devices: private identities and public identities whereas, “Home Network Domain” symbolises the operator network name.

i. **Private Identity:**

The private identity is a unique identifier associated with an individual user or device within a specific domain or network. It is used to identify the user or device within the domain. The private identity is not used for routing of SIP messages, it is used during registration process to authenticate the user.

Private identity is represented in the form of Network Access Identifier (NAI).

ii. **Public Identity:**

The public identity is a unique identifier associated with an individual user or device that can be used to reach them from other domains or networks. It is used for addressing and routing calls from external domains or networks to the intended user or device.

The public identity takes the form of either a SIP URI or a TEL URI. The canonical form of a SIP URI is “sip:username@domain” where “user” represents the username or identifier of the entity, and “domain” represents the domain name.

The canonical form of a TEL URI is “tel:+<CC><NDC><SN>” (max number of digits is 15), that represents an E.164 number.

The SIP URI also supports telephone numbering schemes, which uses E.164 numbering format for telephone numbers. For example, a SIP URI representing a telephone number in E.164 format would look like “sip: +911494354321@domain”, where “+911494354321” is the E.164 telephone number and “domain” is the SIP server's domain name.

iii. **Home Network Domain:**

Operator network is represented using the home network domain name and is in the format of Internet domain name, e.g., operator.com. It may consist of alphanumeric characters separated by a hyphen (-) or dot (.).

8 Numbering, Addressing and domain name in SIP based network.

8.1 Mobility Network

Mobility Network uses wireless communication standard for voice calls using mobile phones and data terminals. In a mobility network, there are two types of user identities available for each user attached i.e Private Identity and Public Identity broadly elaborated as IMPI (IP Multimedia Private Identity) and IMPU (IP Multimedia Public Identity). 3GPP TS 23.003 describes the derivation of these identities which is mentioned below.

Private identity for mobile users is derived from IMSI (International Mobile Subscriber Identity), MCC (Mobile Country Code) and MNC (Mobile Network Code) as “<IMSI>@ims.mnc<MNC>.mcc<MCC>.3gppnetwork.org”. For example, if the IMSI is 234150999999999 (MCC = 234, MNC = 15), the private user identity then takes the form “234150999999999@ims.mnc015.mcc234.3gppnetwork.org”.

Public identity of mobile users is derived from MSISDN (Mobile Station International Subscriber Directory Number), MCC and MNC in sip and tel uri format. For example, if the MSISDN is

+919890000001 (MCC = 234, MNC = 15), the public user identity then takes the form sip:+919890000001@ims.mnc015.mcc234.3gppnetwork.org and tel: +919890000001

Home Network Domain name is derived from MCC and MNC as “mnc<MNC>.mcc<MCC>.3gppnetwork.org”. For e.g., if MCC = 234, MNC = 15, home network domain name would be “mnc.015.mcc.234.3gppnetwork.org”.

A mobile device uses the Home Network Domain name and Private User Identity for registering to IMS network and Public User Identity for making calls.

8.2 Fixed line Network

Historically, most landline telephone services have been delivered over the PSTN. The PSTN has been in place for many decades and generally includes copper wires, and equipment dedicated to supporting call services. Over the last decade, however, telecom service providers have built modern IMS based networks which can support both broadband and landline telephone services.

IMS based network provides multiple benefits like:

- i. Application, network, and device independence: Applications are decoupled from the transport layer. Service providers can deliver common applications and services, with unified authentication, authorization, and accounting across diverse devices in 4G and 5G mobile networks, Wi-Fi networks, fixed networks, and MSO networks.
- ii. Faster delivery of services, at lower cost.
- iii. Enables fixed-mobile convergence by which multiple devices can be used under single subscription.
- iv. Convergence: By evolving legacy TDM voice networks to converged all-IP networks that support voice and IP multimedia services, service providers can reduce OPEX. This helps in reducing TCO (Total Cost of Ownership)
- v. Standards-based solutions: By deploying standards-based network elements, service providers can eliminate vendor lock-in and contain costs.
- vi. Service coverage: IMS standards cover a wide range of IP-based communications services including voice, video, text chat, multiparty conferencing, and collaboration applications.
- vii. Quality of service: IMS supports policy based QoS mechanisms for SLA assurance and to ensure satisfactory user experiences.

8.2.1 Migration of traditional Fixed line Network to IMS based network.

While migrating the fixed line services from traditional PSTN based systems to IMS based systems, it is mandatory to follow the numbering plan and dialling rules as laid by regulatory in the particular country. Additionally, the phone ecosystem needs to evolve to support SIP-based registration and calling functions.

8.2.1.1 Home Domain Name.

A traditional PSTN phone gets registered in Soft Switch, whereas in IMS architecture, IP phone is registered in HSS like any VoLTE number. HSS registration for IP phone is done using Private and Public Identity, which are derived from directory number.

A SIP-based IP phone registers in IMS network using the home domain as the SIP URI, configured under device settings. During registration process, IMS network authenticates the user and creates an

address of record for the specific user. Home domain used by phone is in the format of “user@domain”. Private Identity of the user is used in the authentication and registration process. Once the user is registered to IMS network successfully, it can make outgoing calls and receive incoming calls.

The geographical area of a large country can be divided into multiple regions or provinces with each of the region / province representing a telecom service area. Each of these regions can be represented by a different network name or home domain name to provide telecom services to the users there. In these regions, telecom service providers can offer services customized to the local needs while adhering to local regulations. A distinct network name or home domain name can be applied to users subscribing telecom services from that region. For, e.g., a user subscribing to telecom services from Region1, say, “rg1” can have a home domain as “rg1.wln.opr.ims.com” where “rg1” denotes the Region1, “wln” indicates it’s a fixed line user, “opr” denotes network operator, for e.g., “jio” could be used by Reliance Jio and “ims.com” indicates IMS based services. Thus, all fixed line users from Region1 of Reliance Jio would use the domain “rg1.wln.jio.ims.com” while registering to IMS network and while making calls.

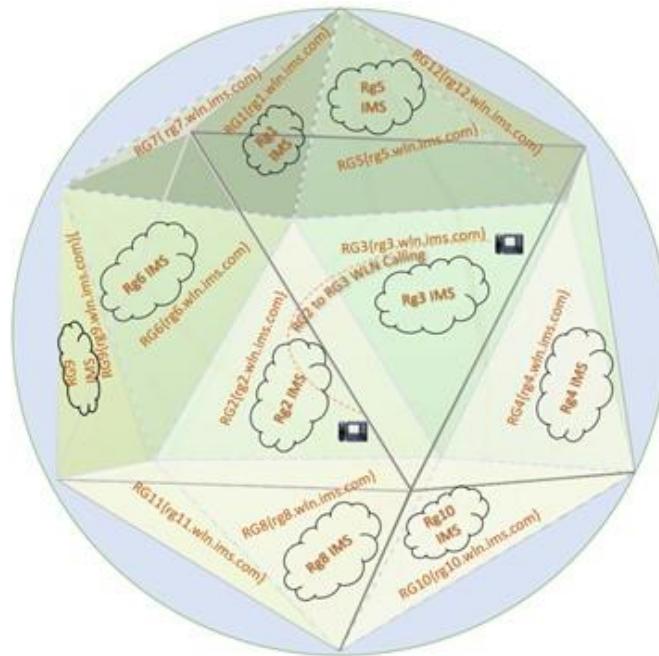


Figure 1. Domain Names across different geographical locations in a large country

8.2.1.2 Private and Public Identity Addressing.

The international ITU-T E.164 specification defines the numbering format for geographic areas and is composed of decimal digits arranged in two code fields: the country code (CC) and the national Significant Number (NSN).

The national significant number may be further subdivided into national destination code and subscriber number fields, according to national requirements.

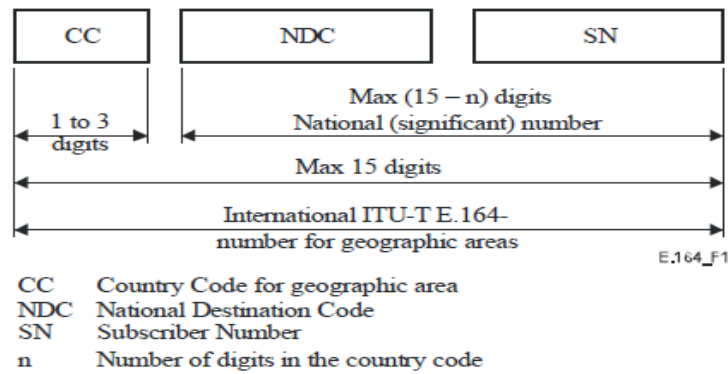


Figure 2 – International ITU-T E.164-number structure for geographic areas

A calling user must dial international prefix “00” if complete E.164 number is used to call a particular user.

In the case of numbers for geographic areas, it is not necessary to always dial the full number including the country code. For instance, it is also possible to dial a:

- local number, consisting solely of the subscriber number, provided the caller is in the same geographical area as defined by the same geographic NDC as the called number.
- national number, consisting of the NSN prefixed by a trunk prefix, for which the digit 0 is recommended by E.164.

So, for example, subject to the caller being in the correct geographical area, a call to the same number could be dialled in the following ways:

Local number: 354321

National number: 01494 354321

International number: 00 91 1494 354321 or +91 1494 354321

It should be noted that in some countries, local number dialling is not permitted, and all calls have to be dialled in national format. Such countries are said to have closed number plans. Local and national numbers are regarded as valid E.164 numbers provided the number can be reached by dialling the number in full international format.

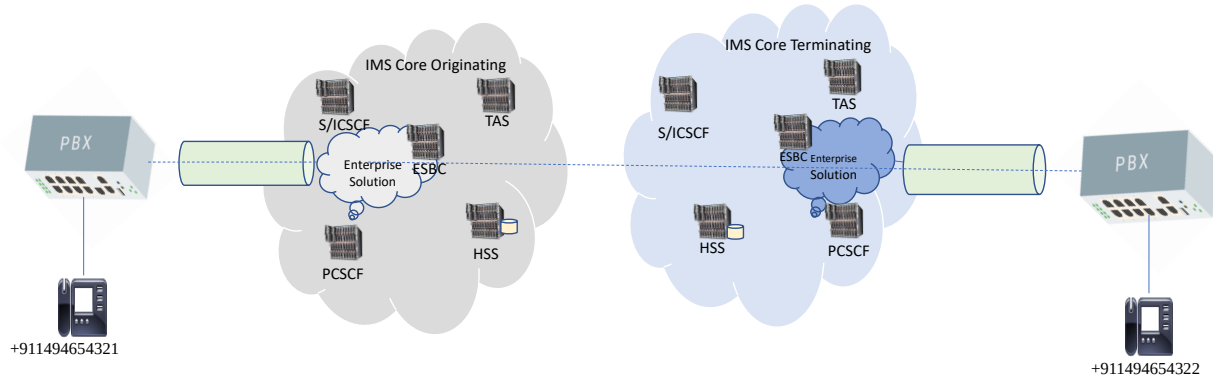
While migrating to IMS based network, the E.164 numbering is still retained, but to process call in IMS, the format username@domain is used. Thus, there is an option to either use alphanumeric identities in the format username@domain or derive the identities using E.164 numbering and domain name.

The retention of E.164 is necessary for the fulfilment of certain requirements, including calling line identity (CLI), emergency calls, lawful interception, and number portability. As a result, private identities and public identities following the E.164 numbering plan would be the best fit solution when migrating to an IMS based network.

The identities thus formed are as shown below:

1. Private Identity: It uses SIP URI format as “sip:<CC><NDC><SN>@home domain”. For example, the private identity of a fixed line number in Region1 “rg1” could be, “sip:+911494354321@rg1.wln.jio.ims.com” where +911494354321 is the actual subscriber number used to make and receive calls and “rg1.wln.jio.ims.com” is the home domain of the subscriber where he/she has subscribed the telephony services from.

2. Public Identity: Both SIP and TEL URI based public identities are defined for the user with SIP URI being the default public identity. The formats of these would look like “sip:+911494354321@rg1.wln.jio.ims.com” and “tel:+911494354321” where “+911494354321” is the actual subscriber number used to make and receive calls and “rg1.wln.jio.ims.com” is the home domain of the subscriber where he/she has subscribed the telephony services from.



During registration process, the SIP phone uses the private identity and home domain, a sample of the same is show below for two numbers:

```
REGISTER sip:rg1.wln.jio.ims.com
From: <+911494354321@rg1.wln.jio.ims.com>;tag=1c511782222
To: +911494354321@rg1.wln.jio.ims.com
Authorization: Digest
username="+911494354321@rg1.wln.jio.ims.com",realm="rg1.wln.jio.ims.com",nonce="",uri="sip:rg1.wln.jio.ims.com",response=""
```

```
REGISTER sip:rg1.wln.jio.ims.com
From: <+911494354322@rg1.wln.jio.ims.com>;tag=1c511782225
To: +911494354322@rg1.wln.jio.ims.com
Authorization: Digest
username="+911494354322@rg1.wln.jio.ims.com",realm="rg1.wln.jio.ims.com",nonce="",uri="sip:rg1.wln.jio.ims.com",response=""
```

Upon successful registration and authentication, IMS network shares the list of public identities to the registered user. For an IP-PBX user, additionally IMS network shares the list of implicitly registered public identities behind the registered pilot number. This facilitates the numbers behind IP-PBX to make/receive calls.

Let's say, the user “+911494354321” calls another fixed line user “+911494354322”.

Calling user's fixed IP phone initiates a call using its public identity in “tel:+911494354321” format in the “From” header. The called party number is populated in “RURI” and “To” header as “tel:+911494354322” which is converted by IMS to “sip:+911494354322@rg1.wln.jio.ims.com” using ENUM query. Further call is processed in IMS based on domain name and routed to called user.

The called user will see the calling party CLI as “+911494354321” and not in the public identity format.

9 Use Cases

As was covered in earlier sections, moving fixed line services to an IMS-based network has numerous advantages. Additionally, this makes it possible to provide fixed line and enterprise subscribers with more value-added services. Below is a list of a few such use cases.

- a. Customized Ring Back Tone (CRBT): Enterprise SIP trunk users can play their own CRBT to the calling party.
- b. Calling Name Display (CNAM): Calling name of enterprises can be displayed to the called user in addition to the calling number.
- c. Fixed Mobile Convergence (FMC): FMC feature enables the customer to receive all incoming calls of a landline to parallelly ring on landline and Mobile enabling the customer to receive calls at his/her convenience. Call can be toggled from fixed line to mobile or mobile to fixed line without disconnecting the call.
- d. Video calling: SIP based IP phones can be offered video calling feature.
- e. One-click conference: List of conference participants can be maintained by devices and when clicked on the list name, network application server will initiate calls to all the participants in the list.
- f. Cloud hosted PBX with multi-level IVR: IP PBX can be hosted in cloud and multi-level IVR feature can be delivered through IMS core application servers.

10 Summary/ proposal:

3GPP defines the methodology to derive home domain name and private and public identities from IMSI, MSISDN, MCC and MNC. On similar lines, the fixed line network can make use of the user directory number to derive the private and public identities while following the E.164 numbering plan. Home domain name for fixed line users can be derived from the different regions within the geographical areas under which service provider provides services. These region names can be concatenated with operator name and wireline service name to form the domain name. Fixed line users can make use of this domain name and private / public identities to register over the IMS network and make/receive calls.

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